

Glossary of Terms

A First Practical Machine Learning Workflow

Retrieval target

You do not need a dictionary definition. You should be able to say each term in plain English and connect it to the notebook work.

Step 01 - Get the Notebook Ready

Term	Say it out loud as...
notebook	The working document where explanation, code cells, and outputs live together.
kernel	The running Python process that executes the notebook cells.
environment	The Python setup and installed packages the notebook is actually using.
package	An installed Python library, such as pandas, NumPy, matplotlib, or scikit-learn.
import	A line of code that makes a package, module, class, or function available.
version number	A printed package or Python version used to confirm what the notebook is running.
Restart Kernel / Run All Cells	The reset habit that checks the notebook works from a clean state.

Step 02 - Create a Tiny Dataset

Term	Say it out loud as...
sample	One example row, such as one house.
feature	An input column the model may use.
target	The known answer column the model will later learn to predict.
DataFrame	A labelled pandas table with rows, columns, and column names.
row	One sample in the table.
column	One measured property or value type in the table.
X	The feature table: the inputs.
y	The target values: the known answers.
shape	The row-and-column size of an object, reported as rows first.

Step 03 - Split Before Training

Term	Say it out loud as...
training data	The rows the model is allowed to learn from.
test data	Held-back rows used to check the trained model later.
train/test split	The act of dividing X and y into training and test portions before fitting.
held-back	Kept away from training so it can be used for a more honest check.
validation set	A development set used to compare choices before the final-ish test check.
test_size	The fraction or count of rows reserved for testing.
random_state	A fixed random seed that makes a split repeatable.
generalise	Behave usefully on examples the model did not train on.

Step 04 - Fit a Regression Model

Term	Say it out loud as...
regression	A machine-learning task where the prediction is a number.
LinearRegression	The scikit-learn model class used for the first regression example.
estimator	scikit-learn's general word for a model-like object.
fit()	Train the model using training features and known training targets.
predict()	Use the fitted model to produce answers for feature rows.
intercept	The baseline part of a linear regression prediction.
coefficient / weight	A learned number attached to a feature.
weighted sum	Features multiplied by weights, then combined with an intercept or bias.
bias	Another common word for the intercept-like offset in a model.

Step 05 - Measure Wrongness: Error, Residual, Loss

Term	Say it out loud as...
prediction	The model's answer for a feature row.
actual value	The known answer used to check the prediction.
residual	actual - predicted; the signed miss.
error	How wrong a prediction is under a chosen rule.
absolute error	The size of the miss, ignoring direction.
squared error	The miss squared, which punishes larger misses more strongly.
mean absolute error (MAE)	The average absolute miss across the evaluated rows.
loss	A chosen rule for turning wrong predictions into a number.
metric	A summary measurement used to report model performance.